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Sub ject : DSA

### Unit 3, S10, SLO1 - Implementation of Queue using Array

**Build the code for Implementation of Queues using Array:**

**a. Declare the queue with size MAX, initialize 'front' and 'rear'**

**b. Write a function enqueue to insert elements at the rear end**

**c. Write a function dequeue to delete elements from the front of the queue**

**Answer:** /\* a. Declaration and Initialization \*/

```
#define MAX 100
```

```
int queue[MAX];
```

```
int front = -1, rear = -1;
```

/\* b. Enqueue Function \*/

```
void enqueue(int val) {
```

```
    if (rear == MAX - 1) {
```

```
        printf("
```

```
Queue Overflow! Cannot enqueue %d\n", val);
```

```
        return;
```

```
    }
```

```
    if (front == -1 && rear == -1) {
```

```
        front = rear = 0;
```

```
    } else {
```

```
        rear++;
```

```
    }
```

```
    queue[rear] = val;
```

```
    printf("
```

```
Enqueued: %d\n", val);
```

```
    }
```

/\* c. Dequeue Function \*/

```
int dequeue() {
```

```
    if (front == -1 || front > rear) {
```

```
        printf("
```

```
Queue Underflow! Queue is empty\n");
```

```
        return -1;
```

```
    }
```

```
    int val = queue[front];
```

```
    front++;
```

```
    if (front > rear) {
```

```
        front = rear = -1; // Reset queue
```

```
    }
```

```
    return val;
```

```
}
```